

J Karthikeyan

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About Me

Computer Science and Business Systems student at Panimalar Engineering College, passionate about building scalable full-stack applications (React, Node.js, MongoDB). Completed 3 industry internships in software development, cloud platforms, and applied AI. Co-authored an IEEE research paper on federated multi-agent orchestration achieving a 57.6% improvement over single-LLM baselines. Certified in Oracle OCI (AI & Foundations), Machine Learning (NPTEL/IIT Kharagpur), and Data Science (NASSCOM Gold). Aspiring to join innovative engineering teams and deliver real-world impact through full-stack development and AI integration.

Education

Panimalar Engineering College, Chennai	CGPA: 8.81 (Up to 5th Sem)	Sept 2023 - Present
B.Tech in Computer Science and Business Systems		
HSLC—Sri Chaitanya School, Hosur	Percentage: 70%	2022 – 2023
SSLC—Mathakondapalli Model School, Hosur	Percentage: 65%	2014 – 2021

Internships

Bluestock Fintech	Jul – Aug 2025
Software Development Engineer Intern (On-site)	
◦ Built and shipped SDE features across 2 production releases; implemented REST APIs, wrote unit tests, and deployed using CI/CD pipeline improving backend reliability.	
ServiceNow University – AICTE / SmartBridge	Oct – Nov 2025
Virtual Internship Program	
◦ Completed ServiceNow Flows, ATF, Administration, and CSA Exam Preparation modules on the Now Platform.	
CSRBOX – IBM SkillsBuild Applied AI – AICTE	Dec 2025 – Jan 2026
Applied AI Virtual Intern	
◦ Built a capstone AI Agent using Python, IBM Watsonx & Relay; covered GenAI, prompt engineering, and agentic automation.	

Projects

- **Spotify Clone (React, Node.js, MongoDB)** 
Full-stack music streaming app with JWT auth, playlist management, and audio playback. ~30% faster load via lazy loading; supports 50+ concurrent sessions. Deployed on Render with GitHub Actions CI/CD pipeline.
- **FinTrack – Personal Finance Tracker (React, Node.js, MongoDB)** 
Finance dashboard with Recharts analytics for income/expense tracking and budgeting. 40% better data visibility; secured with bcrypt + JWT. Containerized with Docker and deployed on Vercel.
- **E-commerce Website (React, Node.js, MongoDB)** 
Full-stack e-commerce platform with JWT auth, secure REST APIs, and unit-tested endpoints (Postman). MongoDB query optimizations reduced read latency by ~25%.
- **Expense Tracker (Python)** 
Python expense tracking system with categorized analytics and automated monthly reporting across 10+ categories using Matplotlib. Tested with pytest for core calculation modules.

Research Publication

- **Decentralized Cognitive Architectures for Autonomous Software Engineering: A Federated Swarm Orchestration Framework for Enterprise-Scale Development Pipelines**
Jeevan C, Karthikeyan J, Lokesh S, Krishna Prasath J, Kishore S — Panimalar Engineering College, Chennai
Proposed FORC, a federated multi-agent framework achieving 85.3% task resolution vs 54.1% single-LLM baseline (57.6% gain) and reducing mean time-to-merge from 4.2 h to 1.8 h on SWE-bench-Enterprise (840 tasks). IEEE Conference Paper (Under Review).

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C, C++, DSA
Frameworks & Libraries: React, Node.js, Express.js, Recharts, Matplotlib
Databases: MongoDB, SQL, Redis (familiar)
DevOps & Cloud: Docker, GitHub Actions, CI/CD, Git, Vercel, Render, Oracle OCI
Testing & Tools: Postman, pytest, Unit Testing, Agile/Scrum, IBM Watsonx, ServiceNow

Additional Certifications

Oracle Cloud Infrastructure 2025 – AI Foundations Associate & Foundations Associate (Aug 2025)
NPTEL Online Certification – Introduction to Machine Learning (IIT Kharagpur, Jul–Sep 2025)
NASSCOM – Data Science for Beginners (Feb 2024, Gold – 89%)
Code-Galatta – Certified in C, C++, Java, Python in DSA (Dec 2023)